

PERSON COUNTY AP, NC, USA

WMO: 722193

Lat: 36.285N

Lon: 78.984W

Elev: 186

StdP: 99.12

Time Zone: -5.00 (NAE)

Period: 04-19

WBAN: 03722

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-7.9	-6.2	-18.8	0.7	-2.5	-16.2	0.9	-1.0	8.9	12.8	8.1	12.4	1.1	310	0.387

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.0	33.9	23.1	32.6	23.0	31.3	22.7	25.3	31.0	24.7	29.9	24.1	28.8	2.4	260

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
23.9	19.2	26.7	23.0	18.1	26.0	22.7	17.8	25.8	78.2	30.8	75.5	30.1	73.4	28.4	27.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
7.5	6.4	5.4	-12.7	35.5	3.5	1.9	-15.2	36.9	-17.3	38.0	-19.3	39.1	-21.9	40.5		
			-13.3	26.0	3.3	0.7	-15.7	26.5	-17.7	26.9	-19.6	27.3	-22.0	27.8		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	14.8	4.2	5.7	9.5	14.6	19.2	23.4	25.1	24.2	21.1	15.0	8.9	6.5	(6)
		DBStd	8.58	5.95	5.35	5.50	4.58	3.84	2.90	2.39	2.68	3.29	4.57	4.84	4.92	
		HDD10.0	649	196	142	78	12	0	0	0	0	9	78	133		
		HDD18.3	2028	438	355	276	128	39	2	0	1	14	123	286	368	
		CDD10.0	2416	17	21	64	149	286	402	467	441	333	166	44	24	
		CDD18.3	751	0	1	3	16	66	154	209	183	97	20	2	1	
		CDH23.3	6708	1	3	43	197	571	1440	1921	1592	751	174	14	2	
		CDH26.7	2505	0	0	4	38	151	562	812	648	251	37	1	0	
(14)	Wind	WSAvg	1.8	2.1	2.3	2.5	2.5	1.9	1.6	1.4	1.2	1.5	1.6	1.7	1.8	(14)
(15)	Precipitation	PrecAvg	1143	88	70	113	89	91	103	114	107	111	87	80	86	(15)
(16)		PrecMax	1524	189	142	207	163	196	305	171	243	429	268	224	196	(16)
(17)		PrecMin	817	29	31	25	27	24	41	65	35	17	1	9	27	(17)
(18)		PrecStd	168	37	27	47	40	41	63	31	45	95	61	53	40	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4% DB	21.3	22.8	27.4	30.2	32.3	36.0	36.2	36.3	33.0	30.2	25.9	22.3		(19)
(20)		0.4% MCWB	15.5	15.2	16.3	17.5	21.1	22.4	23.8	23.6	21.1	21.3	17.1	17.3		(20)
(21)		2% DB	18.8	21.0	24.0	27.6	30.1	32.9	33.7	33.7	31.3	27.3	22.4	19.9		(21)
(22)		2% MCWB	14.5	13.7	14.7	16.8	20.7	22.8	23.7	23.4	21.1	19.7	15.4	16.2		(22)
(23)		5% DB	17.1	17.8	22.2	26.1	27.9	32.0	32.5	32.2	29.8	25.2	20.2	17.5		(23)
(24)		5% MCWB	13.1	12.4	13.9	16.4	20.1	22.7	23.6	23.2	21.5	19.0	15.2	14.7		(24)
(25)		10% DB	13.9	16.0	19.0	23.0	27.1	30.0	31.2	30.2	27.7	22.8	17.8	15.9		(25)
(26)		10% MCWB	9.8	10.4	12.8	15.8	19.4	22.3	23.4	22.9	20.7	18.0	13.6	12.2		(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4% WB	17.4	17.3	18.4	21.8	23.0	25.5	26.0	26.2	24.5	23.5	19.3	19.0		(27)
(28)		0.4% MCDB	18.9	20.6	24.0	26.1	28.4	30.5	31.3	31.2	28.3	26.0	21.5	20.7		(28)
(29)		2% WB	15.6	15.8	16.9	19.5	22.1	24.6	25.4	25.2	23.5	22.1	17.9	17.2		(29)
(30)		2% MCDB	18.2	18.8	20.9	24.1	27.1	29.9	31.1	31.2	27.4	24.7	20.0	18.3		(30)
(31)		5% WB	13.6	13.9	15.7	18.2	21.3	23.9	24.8	24.5	22.7	20.9	16.5	15.6		(31)
(32)		5% MCDB	16.7	18.0	20.0	22.7	25.8	29.1	30.1	29.5	26.6	23.9	19.5	17.7		(32)
(33)		10% WB	10.8	11.0	14.2	17.1	20.6	23.2	24.2	23.8	22.0	19.2	14.2	12.6		(33)
(34)		10% MCDB	13.2	14.7	18.1	21.3	24.7	28.1	29.0	28.1	25.6	22.5	17.5	14.8		(34)
(35)	Mean Daily Temperature Range	MDBR	10.9	11.8	12.4	13.2	11.8	11.6	11.0	10.8	11.0	12.3	12.6	11.0		(35)
(36)		5% DB MCDBR	13.5	14.9	14.7	15.6	13.5	13.3	12.8	13.1	14.1	14.1	15.2	12.9		(36)
(37)		5% DB MCWBR	10.0	9.7	7.7	6.9	5.7	4.6	4.3	4.5	5.2	7.2	9.0	9.8		(37)
(38)		5% WB MCDBR	12.1	13.1	12.5	12.0	11.0	11.7	11.0	10.9	10.3	10.9	11.4	10.6		(38)
(39)		5% WB MCWBR	10.3	9.9	8.0	6.4	5.0	4.6	4.1	4.3	4.6	6.4	8.6	9.4		(39)
(40)	Clear-Sky Solar Irradiance	taub	0.305	0.323	0.355	0.396	0.441	0.487	0.524	0.522	0.412	0.370	0.324	0.309		(40)
(41)		taud	2.513	2.457	2.411	2.297	2.198	2.113	2.029	2.015	2.322	2.422	2.510	2.536		(41)
(42)		Ebn at Noon	895	916	913	887	848	806	772	764	842	850	864	867		(42)
(43)		Edh at Noon	80	95	109	129	145	158	171	170	118	98	80	74		(43)
(44)	All-Sky Solar Radiation	RadAvg	2.52	3.21	4.23	5.32	5.90	6.37	5.97	5.42	4.48	3.64	2.77	2.17		(44)
(45)		RadStd	0.17	0.38	0.30	0.41	0.47	0.45	0.37	0.28	0.41	0.46	0.22	0.19		(45)

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (56 neighbors)	+0.40	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+144	+96

Nomenclature: See separate page